Course Outline
CSE 321: Operating Systems
Semester: Fall 2026

Course Description:
This course provides an introduction to the fundamental concepts, principles, and design of modern operating systems. It emphasizes understanding both the theory and the practical implementation of operating system components, including processes, threads, inter-process communication, CPU scheduling, process synchronization, file systems, memory management, virtual memory, security, and protection. Students learn how an operating system manages processor time, memory, files, storage devices, processes, and other system resources. On the other hand, they use Linux, C programming, shell scripting, POSIX threads, synchronization mechanisms, and the xv6 operating system to implement and analyze different operating system concepts, system calls, process management mechanisms, scheduling operations, and file-system-related features in practice.

Prerequisites: CSE221: Algorithms, CSE340: Computer Architecture

Co-requisites: None

Course Outcomes:

CO1	**Explain** the fundamentals of operating systems and processes, concepts of process management, process states and operations, inter-process communication, system calls, and CPU scheduling mechanisms and algorithms.
CO2	**Apply** multithreading and process synchronization techniques using threads, mutex locks, semaphores, and other synchronization mechanisms to solve race conditions and critical-section problems.
CO3	**Analyze** file and secondary storage management techniques, file system implementation, crash consistency, and journaling, and **evaluate** security and protection mechanisms for controlled, secure access to system resources.
CO4	**Analyze** memory management and virtual memory techniques, including address spaces, paging, page-table structures, demand paging, page faults, and page-replacement algorithms.
CO5	**Implement** and **demonstrate** the functionalities of the operating system through Linux shell commands, shell scripting, C programming language, system calls, process and thread management, synchronization and IPC mechanisms, and the xv6 operating system by **implementing** and **tracing** user programs and system calls, **examining** process structures and scheduler operations, and **developing** file system-related functionalities.

Mapping of CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool:

Sl.	CO Description	POs	Bloom's taxonomy domain /level	Delivery methods and activities	Assessment tools	KPIs of relevant PO	WK (WK1-WK8)	CEP(P1-P7)	SDG	CEA (A1-A5)
CO1	**Explain** the fundamentals of operating systems and processes, concepts of process management, process states and operations, inter-process communication, system calls, and CPU scheduling mechanisms and algorithms.	PO1	Cognitive / Understand	Lectures	Midterm	PO1(d): Apply knowledge of engineering specialization to solve engineering problems.	WK4	N/A	N/A	N/A
CO2	**Apply** multithreading and process synchronization techniques using threads, mutex locks, semaphores, and other synchronization mechanisms to solve race conditions and critical-section problems.	PO1	Cognitive / Apply	Lectures	Midterm	PO1(d): Apply knowledge of engineering specialization to solve engineering problems.		N/A	N/A	N/A
CO3	**Analyze** file and secondary storage management techniques, file system implementation, crash consistency, and journaling, and **evaluate** security and protection mechanisms for controlled, secure access to system resources.	PO 2	Cognitive / Analyze & Evaluate	Lectures	Final	PO2(d): Analyze complex engineering problems using first principles of engineering sciences to reach substantiated conclusions.		P1, P3, P7	N/A	N/A

CO4	**Analyze** memory management and virtual memory techniques, including address spaces, paging, page-table structures, demand paging, page faults, and page-replacement algorithms.	PO 2	Cognitive / Analyze	Lectures	Final	PO2(d): Analyze complex engineering problems using first principles of engineering sciences to reach substantiated conclusions.		P1, P3	N/A	N/A
CO5	**Implement** and **demonstrate** the functionalities of the operating system through Linux shell commands, shell scripting, C programming language, system calls, process and thread management, synchronization and IPC mechanisms, and the xv6 operating system by **implementing** and **tracing** user programs and system calls, **examining** process structures and scheduler operations, and **developing** file system-related functionalities.	PO 5	Cognitive / Apply, Create; Psycho motor / Mechanism	Lectures, Lab	Lab project, lab assignment, lab assessment	PO5(c): Apply modern engineering and IT tools to solve engineering problems.	WK6	P1, P7	N/A	A1, A5

Lesson Plan:

No	Topic	Week/Lecture#	Related CO (if any)
1	OS Fundamentals and Process: a. Process concepts and states b. System calls c. Process management and operations (creation and termination) d. Inter-process communication	Week 1, 2	CO1
2	Threads: a. Multithreading concepts b. Multicore programming c. Multithreading models d. Thread libraries e. Threading issues and operating systems examples	Week 3	CO2
3	CPU Scheduling: a. Concepts of CPU Scheduling b. Scheduling criteria c. Scheduling algorithms and simulations (FCFS, SJF, Priority, Round Robin, and Multilevel queue and Multilevel feedback queue algorithms)	Week 4	CO1
4	Process Synchronization: a. Concepts of process synchronization b. Race condition and critical section problem c. Techniques of process synchronization: Test and Set, Compare and Swap, Mutex locks, Semaphores d. Classical semaphore problems	Week 5, 6	CO2
5	File Systems: a. Magnetic disk structure overview, and magnetic disk architecture b. File concepts, file attributes, and operations c. Indexed File Allocation Method d. UNIX Inode e. File system implementation f. Crash consistency and journaling	Week 7, 8	CO3
6	Memory Management: a. Main memory management background b. Address spaces c. Paging	Week 9, 10	CO4

	d. Page table implementation, Shared pages e. Hierarchical page table f. Introduction of virtual memory g. Demand paging concepts and performance h. Page fault handling i. Page replacement algorithms (FIFO, LRU, Optimal)		
7	Protection: a. Goals and principles of protection b. Domain of protection c. Domain structure and implementation in UNIX and Android d. Access matrix e. Revocation of access rights f. Mandatory access control g. Capability-based systems h. Other protection improvement methods and language-based protection	Week 11	CO3
8	Security: a. Security problem b. Security violation categories and methods c. Security measure levels d. Program threats e. Implementing security defenses f. Firewall	Week 12	CO3

Lab Plan:

Lab	**Topics Covered**	**Lab Activities**	**Submission / Evaluation**
0	Basic Shell Commands, Basic C Programming	Online tutorial(s) provided in the buX of basic Linux shell/bash commands and the fundamentals of the C programming language.	**Evaluation:** Lab Assessment 1 on basic shell commands and C programming.

1	System Calls	Lecture on system calls in C: process manipulation, file/file descriptor manipulation, performance profiling, system call tracing.	**Evaluation:** Lab Assessment 2 on system calls, threads, and synchronization using the C programming language.
2	Linux Shell Scripting	Lecture on shell/bash scripting	N/A
3	Threads and Synchronization	Lecture on creation and management of pthreads and thread synchronization mechanisms using mutex and semaphore.	**Submission:** Lab assignment 1 on threads and synchronization. **Evaluation:** Lab Assessment 2 on system calls, threads, and synchronization using the C programming language.
4	IPC	Lecture on IPC mechanisms such as pipes, shared memory, and message passing.	N/A
5	File Systems	Lab project: Implement features of a custom file system or the file system itself in the C programming language.	**Submission:** Implementation of the project. **Evaluation:** Evaluating the implementation group-wise and individual viva.
6	xv6 - Introduction	Introductory lecture on xv6: introduction, xv6 installation tutorial, booting process, kernel vs user space, file system hierarchy.	N/A
7	xv6 - Features	In-depth explanation: system call implementation and tracing, process structure, context switching, scheduler control.	**Submission:** Lab assignment 2 on the implementation of a scheduler in xv6-RISCV OS.

Lab CO information:

COs	Category	Criteria	Total Marks
CO5	Individual Assignment	Assignment 1	3
CO5	Individual Assignment	Assignment 2	5
CO5	Individual Practical Assessment	Lab Assessment 1	3.5
CO5	Individual Practical Assessment	Lab Assessment 2	3.5
CO5	**Implementation:** Project-Based Group Assessment	Lab Project	10
	Viva: Individual Assessment based on the implementation.		
Total			25

Course Assessment Methods:

Assignments:

Assignments shall be designed to ensure that the students have the required knowledge to analyze and design control systems. Specifically, they will support the student's progress in the project/Laboratory.

Quizzes:

Quizzes will be designed to test the student's understanding of the course and to assess various course outcomes.

Examinations:

The exam shall contain problems designed to test knowledge and comprehension, to analyze control systems, and/or apply the engineering problem-solving method.

Laboratory Work:

The students will gain hands-on experience in designing and analyzing shell commands and scripts on the Linux OS. They will learn how to work with different OS tools, such as system calls, pthreads, and synchronization techniques, and implement schedulers at the OS level. The students will be assessed by multiple lab assignments, lab exams, and a term project.

CO Assessment Plan:

Assessment Tools	Course Outcomes				
	CO1	**CO2**	**CO3**	**CO4**	**CO5**
Midterm	X	X			
Final			X	X	
Lab & Project					X

Textbooks:

1. Operating System Concepts Essentials by Abraham Silberschatz, Peter B. Galvin and Greg Agne, Wiley; 10th Edition (2018). ISBN-13: 978-1119329480
2. Modern Operating Systems by Andrew S. Tanenbaum and Herbert Bos; 5th Edition (2022). ISBN-13: 978-0137618880
3. Operating Systems: Three Easy Pieces by Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau, Arpaci-Dusseau Books, October 2023 (Version 1.10), ISBN-13: 978-1985086593

Course Policies:

Class Policy:

- Classroom and laboratory attendance are mandatory. You should come to the classroom before the instructor. Latecomers may/ may not be allowed to enter the classroom. Students who are absent for more than 30% of theory classes or more than 10% of lab classes will not be allowed to participate in the final examination.
- You should turn off your cellular phone before entering the classroom. You should not leave the classroom to make or take cellular phone calls.
- You should bring a notepad and/or a writing instrument to every class and take detailed notes.
- You should pay attention to the instructor and participate in class discussions.
- You should not do other work during class time.

Honor Code:
Any form of cheating, plagiarism, and/or academic dishonesty will result in an "F" grade in the course.

Late Work and Examinations:
Late assignments will not be accepted. Students who know they will miss class should make arrangements in advance. Exams will be closed-book. There will not be any make-up for quizzes and midterm exams except in cases of hospitalization or detention.

Grading Policies:
Students' grades are assigned according to the grading scale of the BRAC University Undergraduate Study and Examinations Regulations. In addition, the faculty are allowed to take into consideration the class average and standard deviation to reflect the actual class performance for student grade assignments. The grades at the university will be indicated in the following manner:

Marks	Grades
90-100	A (4.0)
85- <90	A- (3.7)
80- <85	B+ (3.3)
75- <80	B (3.0)
70- <75	B- (2.7)
65- <70	C+ (2.3)
60- <65	C (2.0)
57- <60	C- (1.7)
55- <57	D+ (1.3)
52- <55	D (1.0)
50- <52	D- (0.7)
<50	F (0.0)
P	Pass
I	Incomplete
W	Withdrawal
R	Retaken

Course Assessment Methods:
Guidelines for CSE course teaching in BRAC University. The following assessment methods are based on a Theory Course with Lab.

Assessment Tools	**Marks (%)**
1. Assignments	5 %
2. Quizzes/Class Tests	10 %
3. Midterm Examination	25 %
4. Lab & Projects	25 %
5. Final	35 %
Total	**100 %**

CO-wise tentative marks distribution:

CO	**Marks (%)**
CO1	20%
CO2	15%
CO3	20%
CO4	20%
CO5	25%
Total	**100 %**

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